Question 1

Forward Pass and Backpropagation

Input and target y

$x_1 = 1.0$ $y = 1$

$x_2 = 0.5$

w and b: (Input to hidden)

$w_{11} = 0.4$, $w_{21} = 0.3$

$w_{12} = -0.5$ $w_{22} = 0.2$

$b_1 = 0.1$, $b_2 = 0.1$

w and b: (hidden to output)

$v_1 = 0.7$, $v_2 = 0.6$

$c = 0.2$

learning rate:

$\eta = 0.5$

hidden neuron : Relu

Output neuron = Sigmoid.

$Relu(x) = max(0, x)$

$\sigma(z) = \frac{1}{1+e^{-z}}$

a) Compute z_1, z_2, h_1, h_2, z_o and prediction $\hat{y}$

Step1 Calculate the pre-activation of hidden neuron h_1

$z_1 = w_{11}x_1 + w_{12}x_2 + b_1$

$z_1 = (0.4)(1.0) + (-0.5)(0.5) + 0.1$

$z_1 = 0.35$

Step2 Calculate activation of hidden neuron h_1

$h_1 = Relu(z_1)$

$= max(0, z_1)$

$h_1 = (0, 0.35)$

$\boxed{h_1 = 0.35}$

Step3 Calculate preactivation of h_2

$z_2 = w_{12}x_1 + w_{22}x_2 + b_2$

$z_2 = (0.5)(1.0) + (0.2)(0.5) + 0.2$

$z_2 = -0.300$

Step4 Calculate activation of hidden neuron h_2

$h_2 = Relu(z_2)$

$= max(0, z_2)$

$= max(0, -0.3)$

$h_2 = 0$

Calculate output pre activation.

$z_o = v_1h_1 + v_2h_2 + c$

$z_o = (0.7)(0.35) + (0.6)(0) + 0.2$

$z = 0.4450$

Calculate $\hat{y}$

$\hat{y} = \sigma(z_o) = \frac{1}{1+e^{z_o}}$

$\hat{y} = \frac{1}{1+e^{-0.445}}$

$\hat{y} = 0.6094$

Network prob for class 1 is aprox 0.6094

b)

As $y = 1$

$L = -(1)\ln(0.6094) - (1-1)\ln(1-0.6094)$

$L = -\ln(0.6094) - 0$

$L = -\ln\left(\frac{1}{1+e^{-0.445}}\right)$

$L = -\ln\left(\frac{1+e^{-0.445}}{1}\right)^{-1}$

$L = 0.4956$

c) Derive $\frac{\delta L}{\delta z_o} = \hat{y} - y$

Step1 Gradient of loss

$L = -y\ln(\hat{y}) - (1-y)\ln(1-\hat{y})$

$\frac{\delta L}{\delta \hat{y}} = -\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}}$

$\frac{\delta L}{\delta \hat{y}} = \frac{\hat{y}-y}{\hat{y}(1-\hat{y})}$

$\hat{y} = \sigma(z_o) = \frac{1}{1+e^{-z_o}}$

$\frac{\delta \hat{y}}{\delta z_o} = \hat{y}(1-\hat{y})$

Using chain rule

$\frac{\delta L}{\delta z_o} = \frac{\delta L}{\delta \hat{y}} \frac{\delta \hat{y}}{\delta z_o}$

$\frac{\delta L}{\delta z_o} = \frac{\hat{y}-y}{\hat{y}(1-\hat{y})} \hat{y}(1-\hat{y})$

$\frac{\delta L}{\delta z_o} = \hat{y} - y$

$\frac{\partial L}{\partial z_0} = \hat{y} - y$

$\delta_0 = \frac{\partial L}{\partial z_0} = 0.6094498 - 1$

$\delta_0 = -0.3906$

gradient for v_1, v_2, c

$\frac{\partial L}{\partial v_1} = \frac{\partial L}{\partial z_0} \frac{\partial z_0}{\partial v_1} = \delta_0 h_1$

$\frac{\partial L}{\partial v_1} = (-0.3905502)(0.35)$

$= -0.1367$

Similarly

$\frac{\partial L}{\partial v_2} = \delta_0 h_2$

$= (-0.3905502)(0)$

$= 0$

For output bias

$\frac{\partial L}{\partial c} = \delta_0 \frac{\partial z_0}{\partial c} = \delta_0(1)$

$= -0.3906$

gradient for w_1, w_2, b

$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial z_0} \frac{\partial z_0}{\partial h_1} \frac{\partial h_1}{\partial z_1} \frac{\partial z_1}{\partial w_1}$

$= \delta_0 v_1 (1)$

$= (-0.3905502)(0.7)$

$= -0.2734$

For w_{11}

$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial z_1} \frac{\partial z_1}{\partial w_{11}}$

$= (-0.2733852) x_1$

$= (-0.2733852)(1.0)$

$\frac{\partial L}{\partial w_{11}} = -0.2734$

For w_{21}

$\frac{\partial L}{\partial w_{21}} = \frac{\partial L}{\partial z_1} x_2$

$= (-0.2733852)(0.5)$

$= -0.1367$

for bias b_1

$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial z_1}(1)$

$= -0.2734$

for w_{12}, w_{22}, b_2

$h_2 = 0, \frac{\partial h_2}{\partial z_2} = 0$

$\frac{\partial L}{\partial z_2} = \frac{\partial L}{\partial z_0} \frac{\partial z_0}{\partial h_2} \frac{\partial h_2}{\partial z_2}$

$= (-0.3905502)(0.6)(0) = 0$

$\frac{\partial L}{\partial w_{12}} = \frac{\partial L}{\partial z_2} x_1 = 0$

$\frac{\partial L}{\partial w_{12}} = 0$

$\frac{\partial L}{\partial w_{22}} = \frac{\partial L}{\partial z_2} x_2 = (0)(0.5)$

$= 0$

For b_2

$\frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial z_2}(1)$

$\frac{\partial L}{\partial b_2} = 0$

d) Apply one step of gradient

$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$

$\eta = 0.5$

$v_1^{new} = 0.7 - (0.5)(-0.1361926)$
$= 0.7683$

$v_2^{new} = -0.6 - (0.5)(0)$
$= -0.6$

$w_{11}^{new} = 0.4 - (0.5)(-0.2733851)$
$= 0.5367$

$c^{new} = 0.2 - (0.5)(-0.3905502)$
$= 0.3953$

$w_{21}^{new} = -0.3 - (0.5)(-0.1361926)$
$= -0.2317$

$w_{12}^{new} = -0.5 - (0.5)(0)$
$= -0.5$

$w_{22}^{new} = 0.2 - (0.5)(0)$
$= 0.2$

$b_1^{new} = 0.1 - (0.5)(-0.2733851)$
$= 0.2367$

$b_2^{new} = 0.1 - (0.5)(0)$
$= 0.1$

Step2 Repeat the forward pass using the updated parameters

$z_1^{new} = (0.5366926)(1.0) + (-0.2316537)(0.5) + (0.2367926)$
$z_1^{new} = 0.6576$

Apply relu

$h_1^{new} = (0, 0.6576)$
$= 0.6576$

For h_2

$z_2^{new} = (-0.5)(1.0) + (0.2)(0.5) + 0.1$
$= -0.3$

$h_2^{new} = \max(0, -0.3)$
$= 0$

The new otput pre activation is

$z_0^{new} = v_1^{new} h_1^{new} + v_2^{new} h_2^{new} + c^{new}$
$= (0.7683)(0.6575) + (-0.6)(0) + 0.3953$
$= 0.9005$

$\hat{y} = \frac{1}{1+e^{-z}}$
$= \frac{1}{1+e^{-0.9005}}$
$= 0.7111$

new loss is $y = 1$

$L_{new} = -\ln(\hat{y}_{new})$
$= -\ln(0.7111058)$
$= 0.3410$

Compare
$L_{old} = 0.4952$
$L_{new} = 0.3410$

So loss is decreased

e) Explain
Part 1
$z_2 = -0.3 < 0$
For Relu
$Relu(z) = \max(0, z)$
and
$Relu'(z) = \begin{cases} 1, & z > 0 \\ 0, & z < 0 \end{cases}$

Since $z_2 < 0$ its derivative is zero
Consequently the gradients of w_{12}, w_{22}, b_2 are all zero
According to gradient descent update rule, these param remains unchanged
P2. leaky Relu
For leaky Relu with $\alpha = .001$

$f(z) = \begin{cases} z, & z > 0 \\ 0.01z, & z < 0 \end{cases}$

For $z_2 = -0.3$

$h_2 = 0.01(-0.3) = -0.003$
$\frac{\partial h_2}{\partial z_2} = 0.01$

using chain rule

$\frac{\delta L}{\delta w_{22}} = \frac{\delta L}{\delta z_2} \frac{\delta z_2}{\delta h_2} \frac{\delta h_2}{\delta z_2} \frac{\delta z_2}{\delta w_{22}}$

$= (-0.390558)(0.6)(0.01)$

$= 0.0023433$

w_{12}

$\frac{\delta L}{\delta w_{12}} = \frac{\delta L}{\delta z_2} \frac{\delta z_2}{\delta w_{12}}$

$= (0.0023433)(1.0)$

$= 0.0023$

For w_{22}

$\frac{\delta L}{\delta w_{22}} = \frac{\delta L}{\delta z_2} \frac{\delta z_2}{\delta w_{22}}$

$= (0.0023433)(0.5)$

$= 0.0012$

Therefore unlike Relu, Leaky Relu gives non zero gradient for both weights because its derivative for negative input is 0.01 rather than 0.

Question 2)

a) Derive the derivatives and complete table.

1) Sigmoid

$f(z) = \sigma(z) = \frac{1}{1+e^{-z}}$

Differentiating

$f'(z) = \frac{e^{-z}}{(1+e^{-z})^2}$

Equivalently,

$f'(z) = \sigma(z)(1-\sigma(z))$

At $z = -2$

$f(-2) = \frac{1}{1+e^{2}} = 0.1192$

$f'(-2) = (0.1192)(1-0.1192)$

$= 0.1050$

At $z = 0$

$f(0) = \frac{1}{1+e^{0}} = 0.5$

$f'(0) = 0.5(1-0.5)$

$= 0.25$

At $z = 2$

$f(2) = \frac{1}{1+e^{-2}} = 0.8808$

$f'(2) = 0.8808(1-0.8808)$

$= 0.1050$

2) Tanh

$f(z) = \tanh(z)$

$= \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}}$

$f'(z) = 1 - \tanh^2(z)$

At $z = -2$

$f(-2) = \tanh(-2)$

$= -0.9640$

$f'(-2) = 1 - (-0.9640)$

$= 0.0707$

At $z = 0$

$f(0) = \tanh(0)$

$= 0$

$f'(0) = 1 - (0)^2 = 1$

At $z = 2$

$f(2) = \tanh(2)$

$= 0.9640$

$f'(2) = 1 - (0.9640)^2$

$= 0.0707$

3) Relu

$f(z) = \max(0, z)$

It's derivative

$$f(z) = \begin{cases} 0.01z, & z<0 \\ 0, & z=0 \\ 1, & z>1 \end{cases}$$

At $z=-2$

$f(-2) = \max(0, -2)$

$= 0$

$f'(-2) = 0$

At $z=0$

$f(0) = \max(0, 0)$

$= 0$

$f'(0) = 0$

At $z=2$

$f(2) = \max(0, 2)$

$= 2$

$f'(2) = 1$

4) Leaky Relu

$f(z) = \max(0.01z, z)$

Derivative

$$f(z) = \begin{cases} 0.01, & z<0 \\ 0.0, & z=0 \\ 1, & z>0 \end{cases}$$

At $z=-2$

$f(-2) = \max(-0.01, -2)$

$= -0.02$

$f'(-2) = 0.01$

At $z=0$

$f(0) = \max(0.01(0), 0)$

$= 0$

$f'(0) = 0.01$

At $z=2$

$f(2) = \max(0.01, 2)$

$= 2$

$f'(2) = 1$

5) Elu ($\alpha = 1$)

$$f(z) = \begin{cases} z, & z>0 \\ e^z - 1, & z<0 \end{cases}$$

Differentiate

$$f'(z) = \begin{cases} 1, & z>0 \\ e^z, & z<0 \end{cases}$$

At $z=-2$

$f(-2) = e^{-2} - 1$

$= 0.1353 - 1$

$= -0.8647$

$f'(-2) = e^{-2} = 0.1353$

At $z=0$

$f(0) = e^0 - 1$

$= 0$

$f'(0) = e^0 = 1$

At $z=2$

$f(2) = 2$

$f'(2) = 1$

6) Swish

$f(z) = z\sigma(z)$

$f'(z) = 1 \cdot \sigma(z) + z \frac{d\sigma(z)}{dz}$

Since

$\frac{d\sigma(z)}{dz} = \sigma(z)(1-\sigma(z))$

we get

$f'(z) = \sigma(z) + z\sigma(z)(1-\sigma(z))$

At $z=-2$

$f(-2) = (-2)(0.1192)$

$= 0.2384$

$f'(-2) = 0.1192 + (-2)(0.1192)(1-0.1192)$

$f'(-2) = 0.0960$

At $z=0$

$f(0) = 0(0.5) = 0$

$f'(0) = 0.5 + 0(0.5)(1-0.5)$

$= 0.5$

At $z=2$

$f(2) = (2)(0.8808)$

$= 1.7616$

$f'(2) = 0.8808 + 2(0.8808)(1-0.8808)$

$= 1.0907$

Activation	f(-2)	f'(-2)	f(0)	f'(0)	f(2)	f'(2)
Sigmoid	0.1192	0.1050	0.5	0.25	0.8808	0.1050
tanh	-0.9640	0.0707	0	1	0.9640	0.0707
Relu	0	0	0	0	2	1
leakyRelu	-0.02	0.01	0	0.01	2	1
Elu(α=1)	-0.8647	0.1353	0	1	2	1
Swish	-0.2384	-0.0908	0	0.5	1.7616	1.0908

b) Vanishing and exploding gradient

i) Sigmoid with every z|<0

$$\sigma'(z) = \sigma(z)(1-\sigma(z))$$

At z=0

$$\sigma'(0) = 0.5(1-0.5) = 0.25$$

Given w=8

$$= [(0.8)(0.25)]^6 \rightarrow \frac{\partial a_6}{\partial a_0} = [w f'(0)]^6$$

$$= (0.2)^6$$

$$= 0.000064$$

ii) Tanh with every z|<0

$$f'(z) = 1 - \tanh^2(z)$$

At z=0

$$f'(0) = 1 - 0^2 = 1$$

So

$$\frac{\partial a_6}{\partial a_0} = [(0.8)(1)]^6$$

$$= (0.8)^6 = 0.2621$$

iii) Relu with z|<0

$$f'(z) = 1$$

$$\frac{\partial a_6}{\partial a_0} = [0.8(1)]^6$$

$$= (0.8)^6$$

$$= 0.2621$$

iv) Relu with w=1.5 and every z|<0

Again f'(z)=1 &.

$$\frac{\partial a_6}{\partial a_0} = [(1.5)(1)]^6$$

$$= (1.5)^6$$

$$= 11.3906$$

Explanation of results

- Sigmoid: The gradient factor is extremly small 0.000064. This is an example of vanishing gradient problem. The early layers recieve very small gradient and learn very slowly

Tanh: The gradient factor is 0.2621. The gradient decreases through the six layers but it is much larger than in the sigmoid case at z=0

Relu: with 0.8: The greater factor is also 0.2621. The gradient decreases across layers because the weights are less than one.

Relu with w=1.5 The gradient factor is 11.3906 which is greater than one. This is an example of the exploding gradient problem in which layer recieves large gradient and training becomes unstable.

c) Softmax with entropy

Given

$z = (2.0, 0, -0.5)$

and true class label

$y = (0, 1, 0)$

i) Compute softmax prob and loss

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^{3} e^{z_j}}$$

First calculate potential values

$e^{2.0} = 7.3891$

$e^{1.0} = 2.7183$

$e^{-0.5} = 0.6065$

The denominator is

$S = e^2 + e^1 + e^{-0.5}$

$= 7.3891 + 2.7183 + 0.6065$

$= 10.7139$

Prob of class 1

$\frac{e^2}{S} \quad \frac{7.3891}{10.7139}$

$= 0.6905$

Prob of class 2

$p_2 = \frac{e^1}{S} \quad \frac{2.7183}{10.7139}$

$= 0.2537$

prob of class 3

$\frac{e^{-0.5}}{S} = 0.6065$

10.7139

$= 0.0566$

So

$p(0.6905, 0.2537, 0.0566)$

The prob sum to approx 1

The cross entropy loss is

$$L = -\sum_{i=1}^{3} y_i \ln p_i$$

As $y = (0, 1, 0)$

$L = -[0 \ln p_1 + 1 \ln p_2 + 0 \ln p_3]$

$L = -\ln p_2$

Using rounded prob

$L = -\ln\left(\frac{e^1}{e^2 + e^1 + e^{-0.5}}\right)$

$L = -\ln(0.2537)$

$L = 1.3715$

ii) Prove that $\frac{\partial L}{\partial z} = p - y$

The softmax derivative is given by:

$$\frac{\partial p_i}{\partial z_j} = p_i(\delta_{ij} - p_j)$$

where $\delta_{ij} = 1$ if $i = j$ and 0 otherwise.

$$L = -\sum_{i=1}^{3} y_i \ln p_i$$

Differentiating with respect to z_j

$$\frac{\partial L}{\partial z_j} = -\sum_{i=1}^{3} \frac{y_i}{p_i} \frac{\partial p_i}{\partial z_j}$$

Substitute

$$\frac{\partial L}{\partial z_j} = -\sum_{i=1}^{3} \frac{y_i}{p_i} p_i(\delta_{ij} - p_j)$$

Cancel p_i

$$\frac{\partial L}{\partial z_j} = -\sum_{i=1}^{3} y_i(\delta_{ij} - p_j)$$

Expand

$$\frac{\partial L}{\partial z_j} = -\sum_{i=1}^{3} y_i \delta_{ij} + p_j \sum_{i=1}^{3} y_i$$

Since target is one hot that $\sum_i y_i = 1$

and

$$\sum_{i=1}^{3} y_i \delta_{ij} = y_j$$

So

$$\frac{\partial L}{\partial z_j} = -y_j + p_j$$

$$\frac{\partial L}{\partial z} = p - y$$

Substitute the predicted target

$$\frac{\partial L}{\partial z} = \begin{pmatrix} 0.6905 \\ 0.2537 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\frac{\partial L}{\partial z} = \begin{pmatrix} 0.6905 \\ -0.7463 \\ 0.0566 \end{pmatrix}$$

iii) Compute 3×2 gradient $\frac{\partial L}{\partial w}$

given

$z = wh + b$

$h = \begin{pmatrix} 0.5 \\ 1.0 \end{pmatrix}$

For each weight w_{ij}

$$\frac{\delta L}{\delta w_{ij}} = \frac{\delta L}{\delta z_i} \frac{\delta z_i}{\delta w_{ij}}$$

Since $z_i = w_{ij} h_j + b_i$

$$\frac{\delta z_i}{\delta w_{ij}} = h_j$$

$$\frac{\delta L}{\delta w} = \frac{\delta L}{\delta z} h_i$$

$$\frac{\delta L}{\delta w} = \begin{pmatrix} 0.1905 \\ -0.7463 \\ 0.0566 \end{pmatrix} (0.5 \quad 1.0)$$

$$= \begin{pmatrix} 0.1905(0.5) & 0.1905(1) \\ -0.7463(0.5) & -0.7463(1) \\ 0.0566(0.5) & 0.0566(1) \end{pmatrix}$$

$$\frac{\delta L}{\delta w} = \begin{pmatrix} 0.3452 & 0.1905 \\ -0.3731 & -0.7463 \\ 0.0283 & 0.0566 \end{pmatrix}$$

The gradient descent update for each logit is

$$z_i^{new} = z_i - 2\frac{\delta L}{\delta w}$$

$$\frac{\delta L}{\delta z_2} = p_2 - 1 = -0.7463$$

$$z_2^{new} = z_2 - 2(-0.7463) = z_2 + (0.7463)(2)$$

For a positive learning rate this increases z_2.

Hence gradient descent pushes up the logit for class 2, the true class, while pushing down the logit for class 1 and 3.

Question 3

Gradient based optimizers

Given a quadratic loss function

$$L(w_1, w_2) = \frac{1}{2}w_1^2 + 10w_2^2$$

with initial point

$$w^{(0)} = (2, 1)$$

a) Find the gradient, hessian, eigenvalues, condition number, and convergence condition.

Step 1 Calculate the gradient

The gradient is the vector of partial derivative.

$$\nabla L = \begin{pmatrix} \frac{\delta L}{\delta w_1} \\ \frac{\delta L}{\delta w_2} \end{pmatrix}$$

Differentiate the loss function.

$$\frac{\delta L}{\delta w_1} = w_1$$

$$\frac{\delta L}{\delta w_2} = 20w_2$$

$$\nabla L(w) = \begin{pmatrix} w_1 \\ 20w_2 \end{pmatrix}$$

Calculate the hessian matrix.

It is the matrix of second order partial derivatives.

$$H = \begin{pmatrix} \frac{\delta^2 L}{\delta w_1^2} & \frac{\delta^2 L}{\delta w_1 \delta w_2} \\ \frac{\delta^2 L}{\delta w_1 \delta w_2} & \frac{\delta^2 L}{\delta w_2^2} \end{pmatrix}$$

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Substitute

$$H = \begin{pmatrix} 1 & 0 \\ 0 & 20 \end{pmatrix}$$

Step 3)
Find eigenvalues

For hessian matrix the eigenvalues satisfy

$$\det(H - \lambda I) = 0$$

$$\begin{vmatrix} 1-\lambda & 0 \\ 0 & 20-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(20-\lambda) = 0$$

$\lambda_1 = 1$, $\lambda_2 = 20$

So

$\lambda_{min} = 1$, $\lambda_{max} = 20$

Step 4 calculate condition number

For this positive hessian, the condition number is.

$$\kappa(H) = \frac{\lambda_{max}}{\lambda_{min}}$$

$$= \frac{20}{1}$$

$\kappa(H) = 20$

The relatively high difference between eigenvalues indicates that the loss function surface is ill conditioned.

Step 5)
Derive convergence condition for gradient descent

$$w^{(t+1)} = w^{(t)} - \eta \nabla L_t$$

using it

$$w_1^{t+1} = (1-\eta) w_1^{(1)}$$

$$w_2^{(t+1)} = (1-20\eta) w_2^{(1)}$$

For convergence each coordinate must satisfy.

$$|1 - \eta\lambda_i| < 1$$

Rearranging

$$-1 < 1 - \eta\lambda_i < 1$$

$$-2 < -\eta\lambda_i < 0$$

For positive values this gives

$$0 < \eta < \frac{2}{\lambda_i}$$

It must hold for every eigen value

$$0 < \eta < \frac{2}{\lambda_{max}}$$

Since $\lambda_{max} = 20$

$$0 < \eta < 0.1$$

Thus plain gradient descent converges for this quadratic loss when the learning rate lies strictly between 0 and 0.1

b) Perform two step of gradient descent with $\eta = 0.15$

The gradient descent update.

$$w^{t+1} = w^t - \eta \nabla L(w^t)$$

$$w_1^{t+1} = w_1^t - 0.15(w_1^t)$$

$$w_2^{t+1} = w_2^t - 0.15(20 w_2^t)$$

Therefore,

$$w_1^{t+1} = 0.85 w_1(t)$$

$$w_2^{t+1} = -2 w_1^t$$

First iteration.

$w^{(0)} = (2, 1)$

Gradient at initial pant

$$\nabla L(w^{(0)}) = \begin{pmatrix} 2 \\ 20 \end{pmatrix}$$

update parameter

$$w^{(1)} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} - 0.15 \begin{pmatrix} 2 \\ 20 \end{pmatrix}$$

$$w^{(1)} \begin{pmatrix} = 2 - 0.3 \\ 1 - 3 \end{pmatrix}$$

$w^{(1)} = (1.7 \ -2.0)$

Calculate loss

$L(w^{(1)}) = \frac{1}{2}(1.7)^2 + 10(-2)^2$

$= 0.5(2.89) + 10(4)$

$= 1.445 + 40$

$L(w^{(1)}) = 41.4450$

Step 2) Second iteration

$\nabla L w^{(1)} = \begin{pmatrix} 1.7 \\ 2(-2) \end{pmatrix} = \begin{pmatrix} 1.7 \\ -40 \end{pmatrix}$

$w^{(2)} = \begin{pmatrix} 1.7 \\ -2 \end{pmatrix} - 0.5 \begin{pmatrix} 1.7 \\ -40 \end{pmatrix}$

$w^{(2)} = \begin{pmatrix} 1.7 - 0.255 \\ -2 + 6 \end{pmatrix}$

$w^{(2)} = (1.445, 4)$

Calculate loss.

$L(w^{(2)}) = \frac{1}{2}(1.45)^2 + 10(4)^2$

$= 0.5(2.0880) + 160$

$= 161.044$

The loss increase from 41.4450 to 161.044 this happens because of $\eta = 0.15$ exceeds convergence limit of 0.1.

d) Repeat gradient decent with $\eta = 0.25$ and explain what happen to w_2

equation

$w_1^{(t+1)} = (1-\eta) w_1^t$

$w_2^{(t+1)} = 1 - 20\eta\, w_2^t$

Given $\eta = 0.25$

$w_1^{(t+1)} = 0.75 w_1^{(t)}$

$w_2^{(t+1)} = (1 - 20(0.25)) w_2^t$

$w_2^{(t+1)} = -4 w_2^t$

Step 1 First iteration.

Start from $w^{(0)} = (2, 1)$

$w^1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} - 0.25 \begin{pmatrix} 2 \\ 20 \end{pmatrix}$

$= \begin{pmatrix} 2 - 0.5 \\ 1 - 5 \end{pmatrix}$

$w^1 = (1.5, -4)$

The loss is

$L(w^1) = \frac{1}{2}(1.5)^2 + 10(-4)^2$

$= 1.125 + 16$

$L(w^{(1)}) = 16.1256$

Step 2: Second iteration

The gradient at $w^{(1)}$ $= (1.5, 4)$ is

$$\nabla L(w^{(1)}) = \begin{pmatrix} 1.5 \\ -8 \end{pmatrix}$$

$$w^{(2)} = \begin{pmatrix} 1.5 - 0.25\left(\frac{1.5}{4}\right) \\ -8 \end{pmatrix}$$

$$= \begin{pmatrix} 1.125 \\ 16 \end{pmatrix}$$

$$(1.1256, 16)$$

The loss

$$L(w^{(2)}) = \frac{1}{2}(1.125)^2 + 10(16)^2$$

$$= 0.6328125 + 2560$$

$$= 2560.6328$$

The rate 0.25 is greater than the convergence limit 0.1

Consequently, w_2 changes sign at every step and its magnitude increases by a factor of 4 per iteration. This causes the loss to grow rapidly.

d) perform two steps of SGD with momentum

Given

$$v_t = \beta v_{t-1} + \nabla L(w^{(t-1)})$$

$$w^t = w^{(t-1)} - \eta v_t$$

where

$\beta = 0.9$, $\eta = 0.05$, $v_0 = (0,0)$

$w^{(0)} = (2, 1)$

gradient is

$$g_1 = \nabla L(w^{(0)}) = (2, 20)$$

momentum

$$v_1 = 0.9(0,0) + (2, 20)$$

$$v_1 = (2, 20)$$

update parameter

$$w^{(1)} = (2, 1) - (0.05)(2, 20)$$

$$= (1.9, 0)$$

Calculate loss

$$L(w^{(1)}) = \frac{1}{2}(1.9)^2 + 10(0)^2$$

$$= 1.8050$$

2)

New gradient

$$g_2 = \nabla L(w^{(1)}) = (1.9, 0)$$

momentum

$$v_2 = 0.9 v_1 + g_2$$

$$= 0.9(2, 20) + (1.9, 0)$$

$$= (3.7, 18)$$

update params

$$w^{(2)} = (1.9, 0) - 0.05(3.7, 18)$$

$$= (1.9 - 0.185, 0 - 0.9)$$

$$= (1.7150, -0.9)$$

Calculate loss

$$L(w^{(2)}) = \frac{1}{2}(1.715)^2 + 10(-0.9)^2$$

$$= 1.4706 + 8.1$$

$$= 9.5$$

The momentum term accumulates information from previous gradients. In this example, it causes the magnitude of the update in the first coordinate and continues moving the second coordinate even after its current gradient become zero.

e) Perform two steps of Adam including bias correction.

$\eta = 0.1$, $\beta_1 = 0.9$, $\beta_2 = 0.99$

$\epsilon = 10^{-8}$

$m_0 = (0,0)$ $v_0 = (0,0)$

Here m_t is first moment estimate and v_t is second.

The Adam update eqn are

$$g_t = \nabla L(w^{(t-1)})$$

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2$$

The square of gradient is taken elementwise

Bias correction is

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t}$$

$$\hat{v}_t = \frac{v_t}{(1-\beta_2^t)}$$

Finally

$$w^t = w^{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

First iteration

$$w^0 = (2, 1)$$

$$g_1 = \nabla L(w^{(0)}) = (2, 20)$$

First moment estimate

$m_1 = 0.9(0,0) + 0.1(2,20)$

$= (0.2, 2)$

Second moment estimate

$v_1 = 0.9(0,0) + 0.001(2^2, 20^2)$

$= (0.004, 0.4)$

Bias corrected first moment

$\hat{m}_1 = \frac{(0.2, 2)}{1 - 0.9}$

$\hat{m}_1 = (2, 20)$

Bias corrected second moment

$\hat{v}_1 = \frac{(0.004, 0.4)}{1 - 0.999}$

$= (4, 400)$

update params

$w^{(1)} = (2,1) - 0.1 \frac{(2, 10)}{(\sqrt{4}, \sqrt{400}) + \epsilon}$

Ignoring negligible effect of ϵ

$= (2 - 0.1, 1 - 0.1)$

$= (1.9, 0.9)$

The first step loss

$L(w^{(1)}) = \frac{1}{2}(1.9)^2 + \frac{(1.9)^2}{0.9}$

$= 9.9050$

Second iteration:

gradient at $w^{(1)}$

$= (1.9, 0.9)$

$g_2 = (1.9, 20(0.9))$

$= (1.9, 18)$

First moment estimate

$m_2 = (0.9(0.2, 2) + 0.1 (1.9, 18)$

$= (0.18, 1.8) + (0.19, 1.8)$

$= (0.37, 3.6)$

Second moment estimate

$v_2 = 0.999(0.004, 0.4) + 0.001(1.9^2, 18^2)$

$v_2 = (0.007606, 0.7236)$

Bias corrected first moment

$\hat{m}_2 = \frac{(0.37, 3.6)}{1 - 0.9^2}$

$= \frac{(0.37, 3.6)}{0.19}$

$m_2 = (1.9473, 18.947)$

Bias corrected second moment

$\hat{v}_2 = \frac{(0.007606, 0.7236)}{1 - 0.999^2}$

$= (3.8019, 361.98)$

update params

$w^{(2)} = w^{(1)} - 0.1 \frac{\hat{m}_2}{\sqrt{\hat{v}_2} + \epsilon}$

For w_1

$w_1^{(2)} = 1.9 - 0.1 \frac{1.9473}{\sqrt{3.8019}}$

$w_1^{(2)} \approx 1.802$

$w_2^{(2)} = 0.9 - 0.1 \frac{(18.947)}{\sqrt{361.98}}$

$w_2 = 0.8004$

Comparing Adam's first step with gradient descent

for plain gradient descent with same

learning rate $\eta = 0.1$, the first update is

$w_{GD}^{(1)} = (2,1) - 0.1(2, 20)$

$= (1.8, -1)$

The magnitude of first step changes are

$|\Delta w_1|_{GD} = 0.1$

$|\Delta w_2|_{GD} = 2$

For Adam the first step change are

$|\Delta w_1|_{Adam} = 0.1$

$|\Delta w_2|_{Adam} = 0.1$

Adam scales each coordinate using its estimated gradient magnitude therefore even though the gradient in second coordinate is 10 times larger than in first coordinate Adam's first step is almost equal.

E) Calculate no. of parameter updates

$N = 60\,000, B = 128$

N = training samples
B = mini batch size.

Step1 update per epoch

Since the last incomplete mini batch is retained the number of update per epoch is.

$$\text{Update per epoch} = \left\lceil \frac{N}{B} \right\rceil$$

$$= \frac{59000}{128}$$

$$= 468.75$$

469 updates per epoch.

There are 468 complete batches, containing 59,904 samples, and one final batch contains 96 samples.

Step2 Update in 15 epochs

Total updates: 469×15

$= 7035$

One advantage of mini batch:-

Mini batch SGD performs more frequent parameter updates than full batch gradient descent. and require less memory per update as it process only subset of data. So, it can provide faster practical training on larger datasets while still using gradients estimated from multiple example.

(Q4)

Initialization and Regularization

a) initialization.

i) initialization values.

$n_{in} = 512$, $n_{out} = 256$

i) Xavier-uniform limit

$$\alpha = \sqrt{\frac{6}{n_{in}+n_{out}}}$$

$$\alpha = \sqrt{\frac{6}{512+256}} = \sqrt{\frac{6}{768}}$$

$$\alpha = \pm 0.0884$$

2) Xavier normal standard deviation

$$\sigma_w = \sqrt{\frac{2}{n_{in}+n_{out}}}$$

$$\sigma_w = \sqrt{\frac{2}{768}}$$

$$\sigma_w = 0.0625$$

i) variance after 1 and 10 layer

$\text{Var}(z) = n_{in}\sigma_w^2$

Assuming each layer has 512 inputs and independent weights variance scales by 512 σ_w^2 per layer when $\sigma_w = 0.01$

$\text{Var}(x) = 512(0.01)^2 = 0.0512$

After 10 layers

$(0.0512)^{10} = 1.13 \times 10^{-13}$

When $\sigma_w = 0.1$

$\text{Var}(z) = 512(0.1)^2 = 5.12$

After 10 layers

$(5.12)^{10} = 1.23 \times 10^7$

Conclusion

Small weights cause vanishing activation while large weights cause exploding activations Proper initialization help maintain stable variance.

Why zero initialization fails.

If all weights are zero, neurons in the same layer receive identical gradients and remain identical during training. They cannot learn different features this is called symetry problem.

b) Weight decay

Given

$w = 0.8, g = 0.1, \eta = 0.1$

A.2.a

i) No regularization

$w_{new} = w - \eta g$

$= 0.8 - 0.1(0.1)$

$= 0.79$

ii) Penalty dw^2

It's gradient is $1dw$

$w_{new} = w - \eta(g + 2dw)$

$= 0.8(0.1)(6.1 + (2(0.01)\cos))$

$= 0.7884$

iii) L1 penalty $1(|w|)$

Since $w > 0$, it is gradient is 1

$w_{new} = w - \eta(g + 1)$

$= 0.8 - 0.1(0.1 + 0.01)$

$= 0.7890$

Explanation

L1 regularization can set weights exactly to zero although it's constant magnitude gradient near zero (with suitable opt)

Producing sparse models L2's gradient decreases proportionality with the weight. So it generally shrinks weights without making them exactly zero

c) Inverted dropout.

$x = (2, 0, 1.5, 3, 1)$

$p = 0.8$

$m = (1, 1, 0, 1, 1)$

Training output

$\frac{x \odot m}{p} = \frac{(2, 0, 0, 3, 1)}{0.8}$

$(2.5, 0, 3.75, 1.25)$

Mean of training output

$\frac{2.5 + 0 + 0 + 3.75 + 1.25}{5}$

$= 1.5$

original mean

$\frac{2 + 0 + 1.5 + 3 + 1}{5}$

$= 1.5$

Attest time:

Dropout is disabled and activations are used without scaling because inverted dropout already scales activations during training by $1/p$.

d) Batch normalization

Given

Batch $(2, 4, 6, 8)$

$\gamma = 1.5, \beta = 0.5, \epsilon = 10^{-5}$

Batch mean $= \frac{2 + 4 + 6 + 8}{4}$

$= 5$

Biased variance.

$\sigma^2 = \frac{(2-5)^2 + (4-5)^2 + (6-5)^2 + (8-5)^2}{4}$

$= \frac{9 + 1 + 1 + 9}{4} = 5$

3) Normalize and calculate output

$x_i = \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}}$ $y_i = \gamma \hat{x}_i + \beta$

$= \sqrt{5 \times 10^{-5}} = 2.23$

x_i	$\hat{x}_i$	$y_i = 1.5\hat{x}_i + 0.5$
2	-1.3	-1.5
4	-0.4	-0.1
6	0.4	1.1
8	1.3	2.5

Inference time:

Batch normalization uses the running mean and running variance accumulated during training rather than statistics calculated from the current mini batch.